

AI-Powered Customer Support Assistant

Project Overview

Self-Initiated AI Product & Business Analysis Case Study

Prepared by: Anushka Agartaniya Version: 1.0 Date: September 2026

1. Executive Summary

This case study proposes an AI-powered Customer Support Assistant designed to help support agents quickly find relevant and reliable information while resolving customer queries. The proposed assistant uses a Retrieval-Augmented Generation (RAG) approach: it retrieves information from approved organizational knowledge sources and provides that information as context to an AI language model before generating a response.

The solution is designed as an agent-assistance capability rather than an autonomous customer-support system. Support agents remain responsible for reviewing source references, validating AI-generated information, and deciding whether to respond or escalate.

2. Business Problem

Support agents currently rely on multiple knowledge sources, including product documentation, FAQs, troubleshooting guides, and internal support articles. As documentation and customer-query volume increase, finding accurate and relevant information can become time-consuming.

The identified business impacts include:

- Increased response and resolution time.
- Higher support-agent workload.
- Reduced productivity.
- Inconsistent responses.
- Increased escalation of routine queries.
- Potential impact on customer satisfaction.

3. Business Objectives

ID	Objective
BO-001	Reduce time support agents spend searching across knowledge sources for relevant information.
BO-002	Improve speed of responding to customer queries.
BO-003	Improve consistency and accuracy of information provided to customers.
BO-004	Reduce unnecessary escalation of routine customer queries.
BO-005	Improve overall support-agent productivity and customer experience.

4. Users & Stakeholders

Primary user: Support Agent — quickly find accurate information to resolve customer queries.

Business owner: Support Manager — improve support efficiency, response time, and agent productivity.

Product owner: Product Manager — ensure the solution meets business and user needs.

Supporting stakeholders: AI/Engineering, Knowledge Management, Security & Compliance, and Business/Operations teams.

End beneficiary: Customer — receive faster, accurate, and consistent support.

5. Proposed Solution

The proposed solution is an internal AI Customer Support Assistant for support agents. The intended flow is:

- Agent receives a customer query through the existing support process.
- Agent submits the query to the assistant using natural language.
- The assistant processes the query and searches approved, authorized knowledge sources.
- Relevant information is retrieved and supplied as context to the AI language model.
- The assistant generates an AI-assisted response and displays supporting source references.
- The agent reviews and verifies the response before communicating with the customer.
- If reliable information is unavailable, the assistant provides fallback or escalation guidance.
- Agent feedback and usage information support ongoing evaluation and improvement.

6. AI / RAG Approach

The proposed AI approach uses Retrieval-Augmented Generation (RAG). Approved knowledge sources are processed into searchable content. For a support query, the retrieval layer identifies relevant content, which is then provided to the LLM as context for response generation.

The architecture emphasizes:

- Grounding responses in approved organizational knowledge.
- Source transparency so agents can verify supporting information.
- Controlled access to authorized information.
- Safe handling of insufficient or uncertain information.
- Human verification before customer communication.

7. Requirements & Business Analysis Artifacts

The detailed BRD translates the business problem into business requirements, functional and non-functional requirements, risks, personas, user journeys, user stories, acceptance criteria, solution design, AI evaluation, KPIs, and requirements traceability.

Key supporting artifacts:

- User Stories spreadsheet — structured user needs, priorities, functional-requirement links, and acceptance criteria.
- Requirements Traceability Matrix — connects business objectives through requirements, user stories, and acceptance criteria.
- Process Flow — visualizes the proposed support workflow and fallback path.
- RAG Architecture — shows knowledge ingestion, retrieval, generation, source references, and agent verification.

8. AI Evaluation Framework

The proposed evaluation framework combines AI-specific quality measures, operational metrics, human evaluation, and agent feedback. Evaluation should assess whether responses are useful and whether they are supported by approved knowledge, relevant, complete, and appropriately handle uncertainty.

Core evaluation dimensions include:

- Retrieval Relevance
- Groundedness
- Response Relevance
- Response Completeness
- Source Accuracy
- Factual Accuracy
- Hallucination / Unsupported Content
- Uncertainty Handling
- Safety & Access Compliance
- Agent Usefulness

9. Product KPIs & Success Measurement

The BRD defines business, operational, user, and AI-performance metrics. Because this is a self-initiated case study, specific numerical targets are not assumed; baseline values and target improvements should be established during a future pilot using actual support data.

- Average Knowledge Search Time
- Average Response Time
- Average Resolution Time
- Routine Query Escalation Rate
- AI Assistant Adoption Rate
- Query Assistance Rate
- Agent Helpfulness Rate
- Response Groundedness Rate
- Source Reference Accuracy
- Unsupported Response Rate
- Knowledge Gap Rate
- Fallback / Escalation Rate
- Agent Verification Rate
- Repeat Query Rate

10. Key Product & Business Analysis Takeaways

- The proposed assistant is positioned as an AI support-assistance capability, not a replacement for human support agents.
- RAG is used to connect AI-generated responses to approved organizational knowledge.
- Source references and human verification are core trust and safety mechanisms.

- Fallback and escalation are treated as expected product behaviors, not failures to be hidden.
- AI quality should be evaluated separately from agent perception of usefulness.
- Business impact should be measured against a baseline rather than assumed numerical improvements.

Project status: Self-initiated case study. The solution described is proposed and has not been represented as a deployed production system.